# Local & Global Extrema: Which Test, When?

Name _____

**START: find the critical numbers**
Solve $f'(x) = 0$ and find where $f'(x)$ does not exist. Keep only $x$-values in the domain of $f$.

**What is the question asking for?**

## LOCAL MAX / MIN

**Is $f'(c) = 0$ and $f''(c)$ easy to find?**

**YES → 2nd Derivative Test**

- $f''(c) > 0$ → local **MIN** (concave up)
- $f''(c) < 0$ → local **MAX** (concave down)
- $f''(c) = 0$ → **inconclusive**, so use the 1st test

▼ if $f''(c) = 0$

**NO ($f'$ DNE or $f''$ messy) → 1st Derivative Test**

Sign chart of $f'$ on each side of $c$:

- **+ to −** → local **MAX**
- **− to +** → local **MIN**
- **same sign** → neither

## GLOBAL (ABSOLUTE) MAX / MIN

**What is the domain?**

**CLOSED $[a, b]$ → Closed Interval Method**

- Evaluate $f$ at every critical number **inside $[a, b]$ and** at both endpoints.
- Largest value → global **MAX**
- Smallest value → global **MIN**

**OPEN or UNBOUNDED ($(a, b)$ or $\mathbb{R}$) → Sign Chart + End Behavior**

- No endpoints to test (but count any endpoint that is included).
- One critical number: **− to +** → global **MIN**; **+ to −** → global **MAX** there.
- If $f \to \pm\infty$ at an end, that side has none.

## Key details

**CRITICAL NUMBERS**
$c$ is critical if it is in the domain of $f$ and $f'(c) = 0$ **or** $f'(c)$ does not exist (a corner, cusp, or vertical tangent). They are only **candidates**.

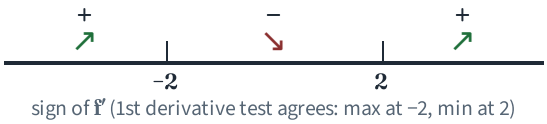
**TRAP: CRITICAL ≠ EXTREMUM**
$f(x) = x^3$ has $f'(0) = 0$ but no max or min. And $f''(c) = 0$ does not mean "no extremum": $x^4$ has a min at 0, $x^3$ has none. Go to the sign chart.

**SAY IT IN WORDS (1ST DERIVATIVE TEST)**
"$f'$ changes from + to − at $x = c$, so $f$ has a local max at $x = c$." Test one $x$ in each interval of the number line; it works at every critical number, even where $f'$ DNE.

**WHERE VS. HOW HIGH**
The extremum **is located at** $x = c$; its **value** is $f(c)$. Report what is asked (location, value, or both). A global extremum can sit at an **endpoint**; on a closed interval and continuous $f$ a global max and min always exist (Extreme Value Theorem).

**WORKED EXAMPLE** $f(x) = x^3 - 12x$

1. $f'(x) = 3x^2 - 12 = 3(x - 2)(x + 2)$, defined everywhere, so critical numbers are $x = -2$ and $x = 2$.
2. **Local:** $f''(x) = 6x$. $f''(-2) = -12 < 0$ → local **max**, value $f(-2) = 16$. $f''(2) = 12 > 0$ → local **min**, value $f(2) = -16$.
3. **Global on $[-3, 5]$:** compare the critical numbers **and** the endpoints (table at right).
4. **Global on $(-\infty, \infty)$:** none. $f \to \infty$ as $x \to \infty$ and $f \to -\infty$ as $x \to -\infty$.



sign of $f'$ (1st derivative test agrees: max at −2, min at 2)

| $x$ | −3 *(end)* | −2 | 2 | 5 *(end)* |
|---|---|---|---|---|
| $f(x)$ | 9 | 16 | **−16** | **65** |

Global max **65** at $x = 5$ (an endpoint, not the local max).
Global min **−16** at $x = 2$.